

Optical Fibers and Cavities

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This study investigates the principles and experimental techniques associated with optical fibers and optical cavities, emphasizing their critical roles in quantum optics laboratories. Efficient coupling of laser beams into multimode and single-mode optical fibers was achieved through systematic beam alignment and mode-matching techniques. Coupling efficiencies of 82.2% and 24.4% for multimode and single-mode fibers, respectively, were initially recorded, with further optimization through a custom-designed two-lens telescope yielding efficiencies of 75.7% and 38.2%. For the optical cavity, despite challenges in achieving measurable photodiode signals due to excessive reflectivity of a back-polished mirror, successful visualization of cavity modes using a CCD sensor confirmed proper alignment and resonance, including TEM_{00} and TEM_{01} modes. These results demonstrate the interplay between theoretical predictions and experimental implementation, providing a framework for optimizing optical systems in quantum optics applications.

I. INTRODUCTION

Due to its unique ability to transmit and distribute light, optical fibers are of great importance in quantum optics laboratories. It can transmit laser beams, photons with minimal loss or unwanted interaction with environment. It is crucial for maintaining the coherence and polarization of quantum states, which are essential for experiments conducted in quantum optics lab.

Coupling is of great importance for fiber optics. Coupling efficiency determines how effectively light can be transmitted into and through the fiber, impacting the overall performance of optical systems. This has importance in both practical application, i.e., communications, and scientific research such as quantum optics lab. In addition to coupling efficiency, fibers are essential for mode shaping and filtering, which are necessary for preparing quantum states or preparing properties for specific experiment practice.

Optical cavities manipulate electromagnetic radiation through precisely engineered resonant structures, it represents a critical frontier in quantum optics research. They are essentially meticulously constructed resonators comprising high-reflectivity dielectric mirrors that can provide experimental platform for investigating fundamental light-matter interactions at the quantum scale. Optical cavities offer a unique mechanism to control, concentrate and coherently manipulate electromagnetic fields, constraining photons within a defined optical path. [1] They enable outstanding experimental capabilities that are unmatched by other experimental setups.

This report presents a theoretical analysis on optical fiber coupling and optical cavity design, along with the experimental execution and result of coupling laser in to multi mode fiber, single mode fiber and optical cavity.

In the second part, we discuss the theory of fiber optics coupling, corresponding experiment procedure, experimental results are analyzed. In third part, we discuss the design of an optical cavity and its construction, experimental results are also analyzed.

Through this investigation, we aim to elucidate the principles and practical applications of both optical fibers and optical cavities, providing valuable insights into their operation and significance in quantum optics laboratories.

II. OPTICAL FIBERS

In this part we first introduce fundamental concepts of optical fiber, then experimental procedure to maximize coupling efficiency for multi-mode fiber and single-mode fiber. Experimental results will be analyzed subsequently.

A. Fundamental Concepts

1. Numerical Aperture

Numerical Aperture is a fundamental parameter characterizing the light guiding capabilities of waveguide structures. It is a dimensionless descriptor that quantifies the maximum angle of optical acceptance and fundamental mode propagation potential within dielectric waveguide systems.[2]

The numerical aperture fundamentally defines the optical collection and transmission efficiency of fiber optic systems, representing a critical interface between geometric optics and wave mechanical principles of light propagation. By characterizing the maximum angle of optical ray acceptance relative to the fiber's optical axis, NA provides a comprehensive metric for understanding

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waveguide optical performance.

The numerical aperture (NA) is derived from the fundamental principles of geometric optics and total internal reflection. Consider a step-index optical fiber with a core of refractive index n_1 and a cladding of refractive index n_2 , where $n_1 > n_2$.

Critical Angle Condition: The condition for total internal reflection occurs when the angle of incidence at the core-cladding interface exceeds the critical angle θ_c : $\sin(\theta_c) = \frac{n_2}{n_1}$

Acceptance Angle Derivation: The maximum angle of acceptance $\theta_{\max}$ is determined by the condition of marginal total internal reflection. At this angle, a light ray is launched at the maximum possible angle that will still be guided within the fiber core.[3] The numerical aperture is defined as: $NA = n_0 \sin(\theta_{\max})$ Where n_0 is the refractive index of the surrounding medium (i.e., air, $n_0 \approx 1$).

Combining Refractive Index Conditions: By combining the critical angle condition with the NA definition: $NA = \sqrt{n_1^2 - n_2^2}$

Numerical Aperture of a Laser Beam: The numerical aperture (NA) of a laser beam represents a critical parameter characterizing the beam's spatial and angular divergence characteristics. Unlike optical fibers, laser NA describes the beam's optical concentration and focusing potential. For a laser beam, the numerical aperture is defined as: $NA = n_0 \sin(\theta_{\text{half}})$ where $\theta_{\text{half}} = \frac{\lambda}{\pi w_0}$

2. ABCD Matrix and q-Parameter

The ABCD matrix technique emerges as a powerful mathematical framework for characterizing electromagnetic wave propagation through complex optical systems. This methodology provides a systematic approach to analyzing ray trajectories and Gaussian beam behavior across diverse optical configurations.

Ray propagation is fundamentally described by two critical parameters, displacement from a reference axis, angular deviation relative to the optical axis. Mathematically, ray transformation can be represented by the matrix equation:

$$\begin{pmatrix} r_2 \\ r'_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r_1 \\ r'_1 \end{pmatrix}$$

Where r represents ray displacement, r' represents ray angle, A , B , C , D are matrix elements characterizing optical elements

Laser beam propagation is described through the complex q-parameter:

$$\frac{1}{q} = \frac{1}{R} - i \frac{\lambda}{\pi w^2}$$

Critical beam parameters include: Beam waist evolution: $w(z) = w_0 \sqrt{1 + (\frac{z}{b})^2}$, Rayleigh range: $b = \frac{\pi w_0^2}{\lambda}$, Beam divergence angle: $\theta = \frac{2\lambda}{\pi w_0}$

The q-parameter transformation through optical systems is governed by:

$$\frac{1}{q_2} = \frac{A \cdot \frac{1}{q_1} + B}{C \cdot \frac{1}{q_1} + D}$$

The ABCD matrix methodology transcends traditional ray tracing by, first providing comprehensive beam characterization, secondly enabling precise optical system design, thirdly quantifying beam waist, divergence, and wavefront curvature

This approach represents a fundamental technique for understanding electromagnetic wave propagation in complex optical systems, bridging geometric and wave optical principles.

B. Experimental Procedure and Result

In this part, we explored the theoretical principles and practical techniques necessary for efficient coupling of light into optical fibers, emphasizing both multi mode and single mode fibers. The lab aimed to build expertise in aligning optical systems and optimizing fiber coupling efficiency through a systematic approach. The result of each part is displayed and analyzed subsequent to the procedure. The fiber we used in this experiment is SM600 from Thorlabs, the specifications[4] are in Table I

1. Beam Waist Measurement

The beam waist measurement was performed prior to coupling and telescope design. This measurement utilized the knife-edge technique, which involves translating a knife edge across the beam and recording the photodiode's output voltage.

TABLE I. Specifications of the fiber used in fiber coupling experiment

Numerical Aperture	Mode Field Diameter
0.10 - 0.14	3.6 - 5.3 μm at 633nm

Data collected displayed in Table II, fitting this data to equation

$$P(x) = \frac{P}{2} \left(1 - \operatorname{erf} \left(\frac{\sqrt{2}x}{w_x} \right) \right)$$

We get Figure 1

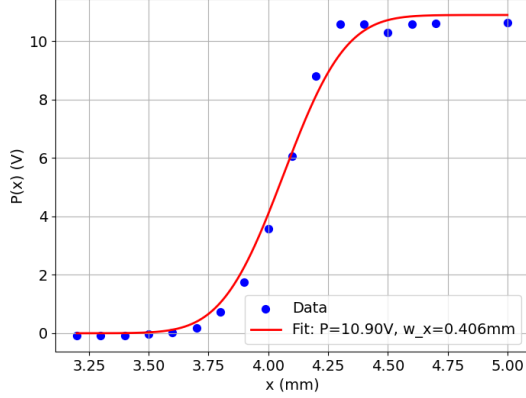


FIG. 1. Data taken and the fitted curve

The beam waist gathered from measurement is 0.406mm.

A minor effect of detector saturation was observed during the experiment. Due to time limitations, the procedure was not repeated. To address this issue in future implementations, the impedance of the terminator was adjusted to prevent saturation. Before adjustment, the impedance is $100k\Omega$, the impedance after adjustment is

TABLE II. Position readings in millimeters (mm) and corresponding photodiode voltage readings in volts (V) obtained during the beam waist measurement.

Position(mm)	Voltage(V)
5.0	10.64
4.7	10.62
4.6	10.59
4.5	10.29
4.4	10.59
4.3	10.58
4.2	8.82
4.1	6.07
4.0	3.58
3.9	1.76
3.8	0.74
3.7	0.18
3.6	0.02
3.5	-0.04
3.4	-0.08
3.3	-0.07
3.2	-0.08

$10k\Omega$ and remained this configuration for the rest of the experiment.

2. Calculating Numerical Aperture

The numerical aperture (NA) of the incident laser can be determined using the following equation:

$$NA = \sin \left(\frac{\lambda}{\pi w_0} \right) \approx \frac{\lambda}{\pi w_0}.$$

The calculated value is $NA_{\text{laser}} = 4.96 \times 10^{-4}$, which is significantly smaller than the NA of the optical fiber. This disparity justifies neglecting the numerical aperture in the evaluation of coupling efficiency.

3. Coupling into Multi Mode Fibers

We began by coupling laser into multi mode fiber to both familiarize ourselves with the alignment process and establish a preliminary alignment that we can later base our coupling into the single mode fiber. Multi mode fibers are simpler to align and more forgiving due to its larger core diameters. It is a reasonable starting point that allow us to have a initial understanding of coupling process. This procedure prove to be rather trivial, data collected can be found in Table III

TABLE III. Coupling efficiency without the telescope for multi-mode and single-mode fibers. Photodiode readouts for before coupling and after coupling respectively

Mode	Efficiency (%)
Multi-mode (Before: 4.10 V, After: 3.37 V)	82.2
Single-mode (Before: 4.10 V, After: 1.00 V)	24.4

TABLE IV. Coupling efficiency with the telescope for multi-mode and single-mode fibers. Photodiode readouts for before coupling and after coupling respectively

Mode	Efficiency (%)
Multi-mode (Before: 3.17 V, After: 2.40 V)	75.7
Single-mode (Before: 3.17 V, After: 1.21 V)	38.2

4. Coupling into Single Mode fiber

After successfully coupling laser into the multi mode fiber, we proceeded to the single-mode fiber. Single mode fibers, with much thinner core diameters, require precise

alignment and matching of input beam's q-parameter. We employed the beam walking technique extensively during this stage, iteratively adjusting the horizontal and vertical alignment of the beam steering mirrors. Small, incremental movements in the mirror angles were alternated to systematically overlap the laser beam with the fiber's core, ensuring maximum power transmission.

Coupling efficiency is displayed in Tabel III

5. Designing and Implimenting Telescope

Next, we designed and constructed a telescope to adapt the free-space laser beam waist to better match the fiber system. The two-lens telescope was calculated and implemented to reshape the beam appropriately for coupling into both multimode and single-mode fibers.

The primary objective of the two-lens telescope is to reduce the beam waist of the incident laser beam prior to its interaction with the collimator. This adjustment ensures that the beam diameter at the output of the collimator is smaller than the mode field diameter (MFD) of the optical fiber, thereby maximizing the coupling efficiency by optimizing the overlap between the incident beam and the guided mode of the fiber.

To achieve this, we apply the equation governing the minimum beam waist produced by a thin lens with a focal length f :

$$w_{02} = \frac{\lambda f}{\pi w_{01}} \frac{1}{\sqrt{1 + \left(\frac{f}{b_0}\right)^2}} = \frac{\lambda f}{\pi w_{01}}$$

Based on the specifications of the single mode fiber and taking into account the finite selection of lenses available in the laboratory, we selected a telescope configuration consisting of a 50 mm eyepiece lens and a 150 mm objective lens. The calculation detail is in Appendix A

Telescope was integrated into the optical system, we repeated the coupling procedures. The telescope enabled us to achieve precise beam shaping, further improving alignment and coupling efficiency. However, we also noticed a significant reduction in intensity for the output of telescope, resulting in effectively reducing the output intensity for multi mode fiber, albeit in proving the coupling efficiency. The specific data is in Table IV

Throughout the process, we utilized photodiode measurements as feedback to optimize coupling. Beam walking played a crucial role in fine-tuning the alignment during each stage, allowing us to achieve peak power transmission by iteratively refining the beam path.



FIG. 2. experimental setup with telescope built-in

6. Output Beam Profile

Due to time constraints, we were unable to investigate the beam profile of the output beam, a theoretical analysis is presented instead.

Light propagates in optical fibers via total internal reflection, with its behavior governed by the refractive index contrast between the fiber core and cladding. While the core guides light, various effects can alter its polarization state. Note that the subject of discussion are all non-polarization maintaining optical fibers

For Single mode fibers, although single mode fibers allow only one mode of light propagation, it does not inherently maintain the polarization of the beam. This is due to birefringence, which arises from imperfections such as asymmetry in the fiber core, mechanical stress, and temperature variations. Birefringence splits the refractive indices into fast and slow axes, causing differential phase delays between orthogonal polarization components.

In terms of transverse modes in single mode fiber, only the fundamental mode, LP_{01} , is supported. This mode has a uniform intensity distribution with no directional dependence, simplifying the analysis of polarization effects. Polarization in single mode fibers is primarily affected by birefringence, which splits the fundamental mode into two orthogonal polarization states aligned with the fiber's fast and slow axes.

In multimode fibers, the situation is more complex due to the presence of multiple spatial modes. Each mode has its own birefringence, leading to mode-dependent polarization effects. Furthermore, mode coupling—the transfer of power between modes—can mix polarization states. This results in a highly dynamic and spatially varying polarization state, making precise control

challenging. The polarization evolution in multi mode fibers requires analyzing the interaction between spatial, temporal, and polarization degrees of freedom.

In terms of transverse mode in multi mode fiber, multiple transverse modes are supported, including higher-order modes such as LP_{11} . Each mode has a distinct spatial field pattern and its own polarization characteristics. The interaction between these modes leads to complex behavior, including modal dispersion, mode coupling, and spatially varying polarization.

In summary, the polarization and transverse mode of output beam does not maintain that of the input beam and are subject to variations caused by disturbance to the fiber.

III. OPTICAL CAVITY

In this part we first introduce the fundamental concepts of optical cavities, then illustrate the detailed experimental procedure and result analysis

A. Fundamental Concepts

1. Linewidth

The line width of a resonance is the frequency full width at half maximum (FWHM) of the power transfer function, in our case, the linewidth is

$$\Delta f = \frac{c}{\pi L} \arcsin\left(\frac{1-r^2}{2r}\right) \approx \frac{(1-r^2)c}{2\pi Lr}$$

The approximation here is valid when $1-r^2 \ll 1$. It is worth noting that L in our set up is the combination of physical arms length, which will be referred to as effective length in the following sections.

2. Free Spectral Range

The *free spectral range* (FSR) of an optical cavity is the frequency spacing between consecutive resonant modes of the cavity. It arises from the interference condition that determines the constructive overlap of light waves within the cavity. Mathematically, the FSR is expressed as:

$$\text{FSR} = \frac{c}{2nL},$$

where c is the speed of light in a vacuum, n is the refractive index of the medium inside the cavity, we can use $n = 1$ in our analysis, and L is the effective length of the cavity.

The FSR represents the periodicity of the cavity's resonant frequencies, indicating the spacing between adjacent longitudinal modes. Physically, it corresponds to the inverse of the round-trip time of light within the cavity, as light travels a distance of $2L$ during one complete round trip.

In practical terms, the FSR sets the scale for the spectral resolution of the cavity and defines the frequency separation at which distinct resonant peaks occur. It is critical in applications such as mode-locked lasers, frequency combs, and optical filters, where precise control of resonant modes is essential. A larger cavity length L results in a smaller FSR, while a higher refractive index n also decreases the FSR by slowing down the effective speed of light.

3. Finesse

The *finesse* of an optical cavity is a measure of its ability to confine light through repeated reflections within the cavity. It is a dimensionless quantity that characterizes the sharpness of the resonance peaks in the cavity's frequency response and provides insight into the cavity's quality factor and energy storage efficiency.

Mathematically, the finesse $\mathcal{F}$ is defined as:

$$\mathcal{F} \equiv \frac{\text{FSR}}{\Delta f} \approx \frac{\pi r}{1-r^2}$$

4. Quality

The Quality factor (Q-factor) of a resonance is the ratio of the resonant frequency to the linewidth,

$$Q_m \equiv \frac{f_m}{\Delta f}$$

A higher Q-factor corresponds to a sharper resonance peak, indicating that the system retains energy for a longer time relative to the oscillation period. A lower Q-factor signifies broader resonance peaks and faster energy dissipation.

B. Cavity Design and Alignment

In this experiment we use a cavity consisting of one concave mirror and two backside polished mirrors for input and output, one regular mirror that is piezo attached.

Now that we've established the choice of mirrors, we should determine the spacing of the mirrors.

In order to make construction and calculation simpler, we decided to make the layout in rectangular shape, enabling the distance from the input coupling mirror to the concave mirror and the distance from the curved mirror back to the input coupling mirror the same, i.e., the effective length mentioned earlier. Enabling simpler calculation, construction and varification.

The design of our cavity primarily focuses on the stability of the cavity.

1. Stability of Cavity

For the cavity to be stable, its ABCD matrix elements should satisfy the following relation: $\frac{(A+D)^2}{4} = (2\eta - 1)^2$, Where $0 \leq \eta \leq 1$. The ABCD matrix of this optical cavity can be calculated as:

$$M = \begin{pmatrix} 1 - \frac{2L}{R_c} & 2L - \frac{2L^2}{R_c} \\ -\frac{2}{R_c} & 1 - \frac{2L}{R_c} \end{pmatrix}.$$

The stability relation is then

$$(2\eta - 1)^2 = \frac{(1 - \frac{2L}{R_c} + 1 - \frac{2L}{R_c})^2}{4} = \frac{(2 - \frac{4L}{R_c})^2}{4} = (\frac{2L}{R_c} - 1)^2$$

Therefore in order for our cavity to be stable, the following inequality must stand $0 \leq \frac{1}{2}(\frac{2L}{R_c}) \leq 1$

2. Size of w_0 and its Position

Using q parameter $q_{res} = z_0 + ib$ where z_0 is the position of minimum waist and $b = \frac{\pi w_0^2}{\lambda}$

$$q_{res} = \frac{A - D \pm \sqrt{(A + D)^2 - 4}}{2C} = i \frac{\sqrt{2L(2R_c - 2L)}}{2}$$

we can derive the location and beam waist as follow:

$$z_0 = 0, w_0 = \sqrt{\frac{\lambda b}{\pi}} = \sqrt{\frac{\lambda \sqrt{2L(2R_c - 2L)}}{2\pi}}$$

with our parameters, $L = 0.323m$ (i.e., the sum of the diagonal and longer side of a three by six inches rectangle), $R_c = 2f_c = 1m$, $\lambda = 633nm$ here f_c is the focal length of concave mirror in the cavity. The Beam waist can be calculated as $w_0 = 0.132mm$

3. Cavity Alignment

The alignment of an optical cavity is a nuanced process that demands precision and systematic approach. The initial phase of input coupling and mirror alignment begins by strategically positioning the input mirror to

create optimal beam propagation. This involves carefully overlapping the incoming beam with the cavity's internal light path, a process that requires meticulous adjustments of mirror orientations.

The fine-tuning process utilizes a technique known as "beam walking," wherein two strategically selected mirrors, namely the output mirror and the one diagonal to it, are iteratively adjusted in both horizontal and vertical planes. This method involves making minute, incremental movements, closely observing the resulting optical response. When alignment begins to deteriorate, we must systematically reverse the direction of mirror adjustments.

Mode matching represents a sophisticated stage of cavity alignment that transforms the input beam's characteristics to optimally couple with the cavity's fundamental mode. This process involves designing a telescope system using carefully selected lenses to manipulate the beam's q-parameter. Typically, we would employ a two-lens configuration, with one lens often mounted on a precision translation stage to enable fine-tuned adjustments. However, due to time limitations, we were unable to perform mode matching in this experiment.

C. Experimental Results

1. Photodiode Readout and Analysis

Our experimental setup failed to collect any valid photodiode output, here's our analysis on why this happened.

We performed extensive investigation on why our cavity failed to give out valid output.

We first examined the mirror type and product number to ensure no trivial error exist in our configuration.

Subsequent to that, we checked the laser beams in the cavity to make sure that the resonance is valid and that this cavity is well aligned.

Furthermore, the mode pattern, will be presented in the next section, of the out coupling laser gathered by CCD sensor indicate high level of alignment and resonance. Giving more evidence to the error not being the construction or alignment.

By this point we've eliminated error in cavity construction, the probable cause remained is primarily the equipment not performing as expected. The photodiode is examined by shining nontrivial amount of light into it and examine the oscilloscope, which resulted in

reasonable output. Meaning the detector system is reliable.

By performing the above procedure, we've narrow the possible error down to the mirrors themselves, namely the input and output mirrors might be excessively reflective. To prove this theory, we first removed the input mirror and let incident beam incident directly on output coupling mirror, the transmitted beam is sufficiently powerful. We then mounted the input coupling mirror back on and dismounted the output coupling mirror and discovered the exiting beam of this back polished mirror is extremely dim. We attempted to measure the output power of this input coupling mirror and failed to gather any reading from the detector system. Indicating the error resides in the input coupling mirror.

The increased reflectiveness of a back-polished mirror can be influenced by several external factors, primarily stemming from the optical environment and substrate interactions. One major contributor is the refractive index contrast between the mirror substrate and the surrounding medium. For example, when the mirror is surrounded by a low-refractive-index medium, such as air, Fresnel reflections at the back-polished surface become more pronounced, increasing the contribution of back reflections to the overall reflectivity. Furthermore, external light sources can pass through the front coating and reflect off the polished back surface, re-entering the system and enhancing the apparent reflectivity. Environmental factors, such as temperature fluctuations, can also alter the refractive index of the surrounding medium or the substrate itself, affecting the back-surface reflectance. Additionally, contaminants or dust on the back-polished surface can scatter light in unpredictable directions, potentially redirecting light that might otherwise escape, thereby increasing the observed reflectivity. These external factors demonstrate how environmental and material properties surrounding a back-polished mirror can amplify its reflective behavior, independent of the alignment within the optical system.

For the purpose of discussion, the procedure for analyzing oscilloscope reading is described in Appendix B

2. Visualizing Cavity Modes

We visualized cavity modes by collecting output mode as video files. To achieve this, we setup the CCD sensor provided by Thorlabs. Although no reasonable data can be collected from our setup using photodiode detectors. The CCD sensor is less prone to noise and is more sensitive. We managed to receive sufficiently identifiable mode patterns. The sensor translates the intensity of the incident beam to digital signal in the form of image or video, allowing direct analysis on the mode itself.

To interpret the mode pattern of an out-coupled laser from an optical cavity, analyze the spatial intensity distribution to identify the transverse electromagnetic modes (TEM_{mn}) supported by the cavity. A Gaussian intensity profile indicates the fundamental TEM_{00} mode, the corresponding mode pattern was identified in our experiment and depicted in Figure 4, characterized by high coupling efficiency and optimal alignment. Patterns with distinct nodes or lobes, such as TEM_{10} or TEM_{01} , identified and presented in Figure 3, suggest higher-order modes, typically caused by beam misalignment, mismatched mode waists, or cavity imperfections. Symmetry in the mode pattern reflects proper alignment and stable cavity operation, while irregular or distorted patterns may indicate mechanical vibrations, thermal effects, or optical defects. Advanced diagnostics, such as beam profiling and Fourier analysis, can help decompose complex mode patterns and evaluate the contributions of multiple modes.



FIG. 3. $TEM_{0,1}$ exiting the cavity

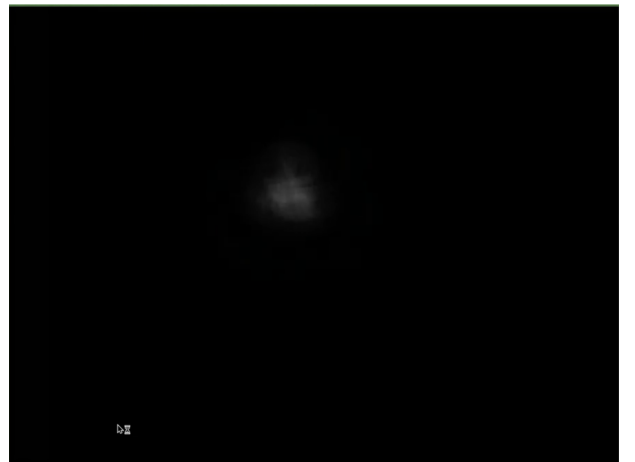


FIG. 4. $TEM_{0,0}$ exiting the cavity

IV. CONCLUSION

This lab delved into the fundamental principles of optical fiber and optical cavities, specifically focusing on the coupling and mode matching. The optical fiber was meticulously coupled with laser and fine tuned to achieve optimal efficiency, output laser power was measured using photodiode and efficiency was analyzed using the readouts of photodiode. Optical cavity was carefully constructed and optimized, mode pattern was gathered by sensor and analyzed.

Furthermore, the experiment explored the possible cause for low transmission efficiency for optical cavity. Extensive analysis was conducted to investigate different aspect of the cavity.

This experiment established a reliable methodology for fiber coupling validation and provided profound insights into the characteristics of the optical cavity.

Appendix A: Telescope Design

The purpose of using a telescope in coupling laser into optic fibers is to match the incident beam waist to MFD of the fiber, hence improve the coupling efficiency. As discussed earlier, the incident laser has a beam waist of 0.406mm. The MFD of the Single mode fiber we used, Thorlabs SM 600 is between $3.6\mu\text{m}$ and $5.3\mu\text{m}$ [4] by applying this python script below with lenses at hand, we determined the best choice for us is 50mm for the eyepiece lens and 150mm for the objective lens.

Listing 1. Code used to design telescope

```

1  import numpy as np
2
3  import numpy as np
4
5  def Gaussian_Beams(R, WaveLength, waist):
6      """Gaussian Beam discribed by 1/q, returns
7          ↪ 1/q"""
8      return 1/R - 1j*WaveLength/(np.pi*waist
9          ↪ **2)
10
11 def Transformation(q_r, TransformMatrix):
12     """How the reciprocal of q evolve along
13         ↪ the way, returns 1/q' """
14     return (TransformMatrix[1,0]+
15         ↪ TransformMatrix[1,1]*q_r)/(
16         ↪ TransformMatrix[0,0]+
17         ↪ TransformMatrix[0,1]*q_r)
18
19 def Radius(q_r):
20     """Calculate curvature radius, returns
21         ↪ curvature radius"""
22     return 1 / np.real(q_r)
23
24 def Waist(q_r, WaveLength):
25     """Calculate waist, returns waist"""
26     return np.sqrt(-1*WaveLength/(np.imag(q_r)
27         ↪ *np.pi))

```

```

20
21 # set parameters
22 R_in = np.inf
23 waist_in = 0.4e-3
24 WaveLength = 633e-9
25 f_e = 0.10
26 f_o = 0.10
27 TransformMatrix = np.array([[ -1*f_e/f_o, f_e +
28     ↪ f_o],
29                             [      0,      -1*f_o/
30     ↪ f_e]])
31
32 # calculate initial q in reciprocal
33 q_r_in = Gaussian_Beams(R_in, WaveLength,
34     ↪ waist_in)
35
36 # How the reciprocal of q evolve along the way
37 q_r_out = Transformation(q_r_in,
38     ↪ TransformMatrix)
39
40 # result radius
41 R_out = Radius(q_r_out)
42
43 # result waist
44 waist_out = Waist(q_r_out, WaveLength)
45
46 print(" the curvature radius:", R_out,"m\n", "
47     ↪ the waist:" , waist_out, "m")

```

Appendix B: Analysis on Cavity Properties Using Photodiode Readout

To analyze the finesse and linewidth of a bowtie optical cavity using the peaks observed on an oscilloscope, we need to first understand the origin of the signal. The peaks arise due to the resonance condition of the cavity, which is modulated by the periodic displacement of the back-polished mirror driven by the PZT. The oscilloscope displays these resonance peaks as the PZT varies the cavity length, enabling precise measurements of the cavity's frequency response.

The first step in the analysis is to determine the free spectral range (FSR) of the cavity. This is achieved by measuring the time difference between two peaks, which corresponds to the time interval for the cavity length to sweep through one full wavelength. Assuming the PZT drive produces a linear change in cavity length over time, which is what we need and achieved by applying triangular waveform input, the free spectral range in terms of frequency can be calculated as the reciprocal of the time interval between adjacent peaks.

The second step involves determining the full width at half maximum (FWHM) of a single resonance peak. The FWHM represents the frequency width of the resonance and can be extracted by identifying the time positions at which the peak amplitude falls to half of its maximum value. The time width is then converted to frequency width using the PZT's calibrated modulation rate.

Once the FSR and FWHM are determined, the finesse of the optical cavity can be calculated. The finesse, defined as the ratio of the free spectral range to the FWHM, provides a measure of the cavity's spectral resolution and

is a direct indicator of its optical quality. The linewidth of the cavity resonance is equivalent to the FWHM in terms of frequency.

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